UNIVERSITÉ D'OTTAWA • UNIVERSITY OF OTTAWA
Sciences • Science
Mathématiques et statistique
Mathematics and Statistics

MAT 2355 Mid-term Exam

February 27, 2001. Duration: 80 minutes

Family Name:_____

First Name:_____

Student number:_____

1	
2	
3	
4	
(Bonus) 5	
Total	

PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE BEGINNING:

1. You have 80 minutes to complete this exam.

2. This is a closed book exam, and no notes of any kind are permitted. Calculators are not allowed, and the use of cell phones is not permitted.

3. The correct answer requires reasonable justification written legibly and logically –you must convince me that you know why your solution is correct.

4. Each question is worth an equal numbr of points. Questions 1 to 4 should be attempted by everyone. Question 5 is a bonus question, and should bot be tried until all parts of questions 1-4 have been completed and checked.

5. Please use the space provided, including the backs of pages if necessary. If you need scrap paper, please ask.

1. Let $f : \mathbf{R}^2 \to \mathbf{R}^2$ be defined by $f(x, y) = \frac{1}{5}(3x + 4y + 5, 4x - 3y)$.

   a) Show that f is an isometry.

   b) Is f orientation preserving or orientation reversing?

   c) Is f a reflection? If so, find the line of reflection, and if not, give reasons why it is not.

2.

a) Express the translation of $\mathbf{R}^3$ defined by $f(x, y, z) = (x - 2, y, z + 2)$ as a product of two reflections.

b) Express the rotation by $\frac{\pi}{2}$ about $(1, 1)$ as a product of two reflections. Check your answer.

c) (i) Show that if $P = (0, 0)$, $Q = (1, 1), R = (1, -1)$ and $S = (2, -2)$, then there is an orientation preserving isometry $f : \mathbf{R}^2 \to \mathbf{R}^2$ such that $f(\overline{PQ}) = \overline{RS}$. Check your answer.

(ii) Find an orientation *reversing* isometry g, such that $g(\overline{PQ}) = \overline{RS}$.

3. Let $H_1 = \{v \in \mathbf{R}^3 \mid (1,0,-1) \cdot v = 0\}$ and $H_2 = \{v \in \mathbf{R}^3 \mid (1,0,1) \cdot v = 1\}$.

a) Find formulae for the reflections R_{H_1} and R_{H_2} in H_1 and H_2 respectively.

b) Use your answers from (a) to find $A \in O(3)$ and $b \in \mathbf{R}^3$ so that $R_{H_2} \circ R_{H_1} = Av + b$.

c) Show that the function $g(v) = Av$ is the rotation by π about the $y-$axis. (You may use the formula $R_{a,\theta}(v) = (v \cdot a)(1 - \cos\theta)a + (\cos\theta)v + (\sin\theta)a \times v$.)

d) Using (c), or otherwise, show that $R_{H_2} \circ R_{H_1}$ is the rotation by π about the line $L = \{(\frac{1}{2}, s, \frac{1}{2}) \mid s \in \mathbf{R}\}$.

4. Let $g : \mathbf{R}^3 \to \mathbf{R}^3$ be defined by $g(v) = Av$, where $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}$.

a) Show that g is an orientation reversing isometry.

b) Find the set of fixed points of g.

c) Show that g is not a reflection.

5. (Bonus: Do not attempt this question until you have completed all parts of questions 1-4)

a) Recall the isometry of question 4, namely $g : \mathbf{R}^3 \to \mathbf{R}^3$ defined by

$$g(v) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} v.$$

Find a vector $0 \neq a \in \mathbf{R}^3$ and $\theta \in [0, 2\pi)$ so that g is rotation by θ about the directed line through the oirgin with direction vector a, followed by reflection in the plane through through the origin with normal vector a.

b) Recall the isometry of question 1, namely, $f : \mathbf{R}^2 \to \mathbf{R}^2$ defined by

$$f(x, y) = \frac{1}{5}(3x + 4y + 5, 4x - 3y).$$

Find a line $L \subset \mathbf{R}^2$, and a vector w, parallel to L, so that f is translation by w followed by reflection in L.